

Sustainability aware asset management

Lecture 9: Temperature Alignment and Net Zero Investment

Eric Jondeau

Faculté des HEC, Université de Lausanne
Master of Science in Finance

- **Bolton, P., M. Kacperczyk, and F. Samama** (2021), *Net-Zero Carbon Portfolio Alignment*. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3922686
- **Cheng G., Jondeau E., and B. Mojon** (2023), *Building Portfolios of Sovereign Securities with Decreasing Carbon Footprints*:
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4207316
- **Institut Louis Bachelier et al.** (2020), *The Alignment Cookbook - A Technical Review of Methodologies Assessing a Portfolio's Alignment with Low-carbon Trajectories or Temperature Goal*:
<https://www.louisbachelier.org/wp-content/uploads/2020/10/cookbook.pdf>
- **Jondeau E., B. Mojon, and L.A. Pereira da Silva** (2021), *Building Benchmarks Portfolios with Decreasing Carbon Footprints*:
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3987186

Objectives of the Lecture

- **Temperature Alignment and Net Zero Investment**
- Exclusion/Best-in-class Strategies
- Minimization of the Tracking Error
- Corporate Bonds
- Sovereign Bonds

Portfolio Temperature Alignment

Paris Agreement (2015): hold the increase in the global average temperature to well below 2°C above preindustrial levels, and pursue efforts to limit the temperature increase to 1.5°C.

- Limit GHG emissions down to a “net-zero” level.
- The sooner we reach zero, the closer we will be to the +1.5°C limit.

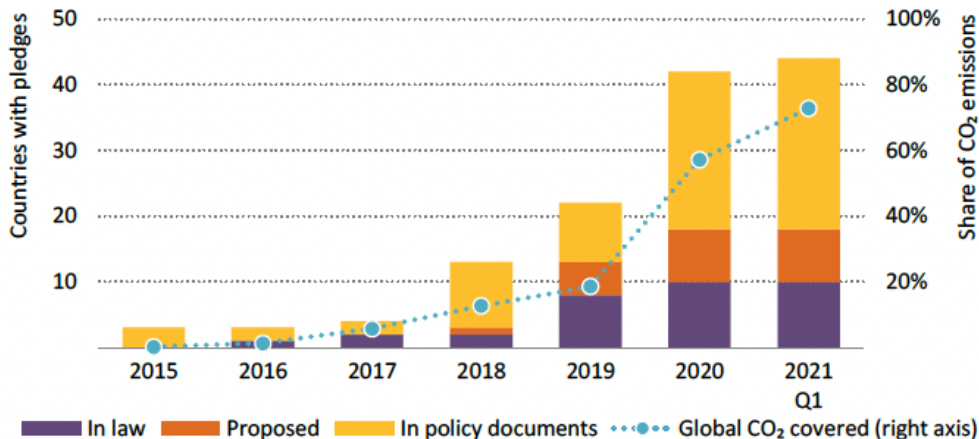
Countries: ~ 130 countries target to be carbon neutral by 2050 (China by 2060).

Central banks and supervisors: **Network for Greening the Financial System (NGFS)** – strengthen the global response to meet the Paris Agreement goals (90 members and 14 observers)

Private sector financial institutions: **Asset Owners Net-Zero Alliance** – 37 institutional investors (~ \$5.7 trillion) have committed to align their portfolios with a 1.5°C consistent trajectory by setting science-based targets by 2050.

National Net Zero Pledges

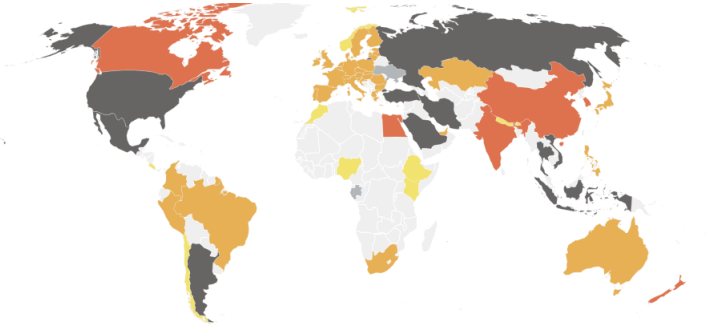
Number of national net zero pledges and share of global CO₂ emissions covered



Source: IEA (2021), *Net Zero by 2050: A Roadmap for the Global Energy Sector*

National Net Zero Pledges

Overview of Climate Action Tracker's net zero target evaluations



Assessments of:



AVIATION

[Find out more](#)



SHIPPING

[Find out more](#)

The maps displayed are for reference only.

LAST UPDATE: December 2025

CRITICALLY INSUFFICIENT

HIGHLY INSUFFICIENT

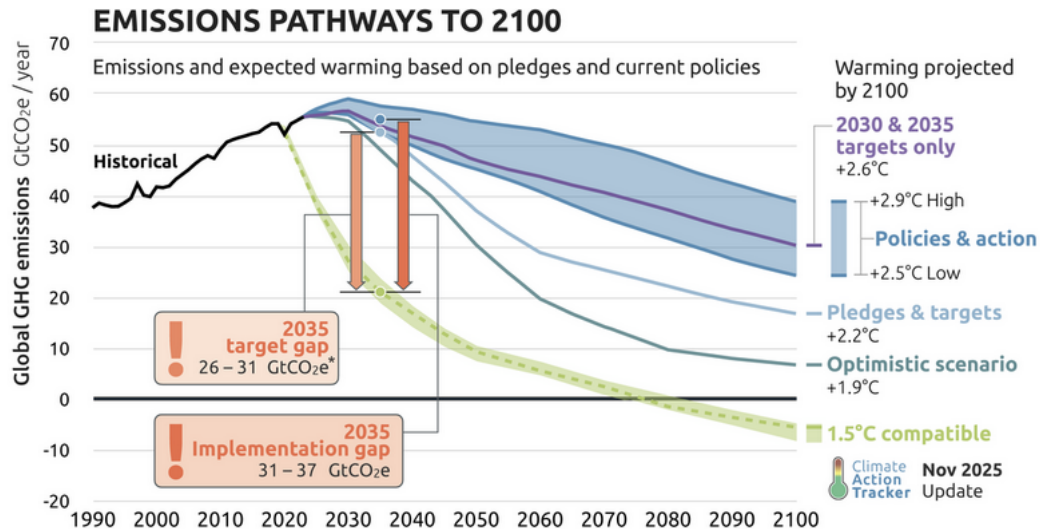
INSUFFICIENT

ALMOST SUFFICIENT

1.5°C PARIS AGREEMENT
COMPATIBLE

Source: <https://climateactiontracker.org/>

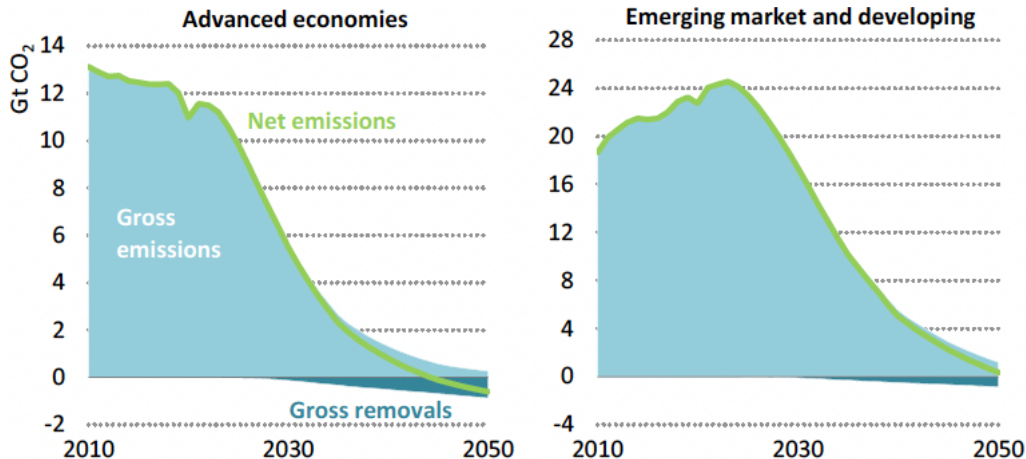
National Net Zero Pledges



Source: <https://climateactiontracker.org/global/emissions-pathways/>

CO₂ Emissions in the Net-Zero Emission Scenario

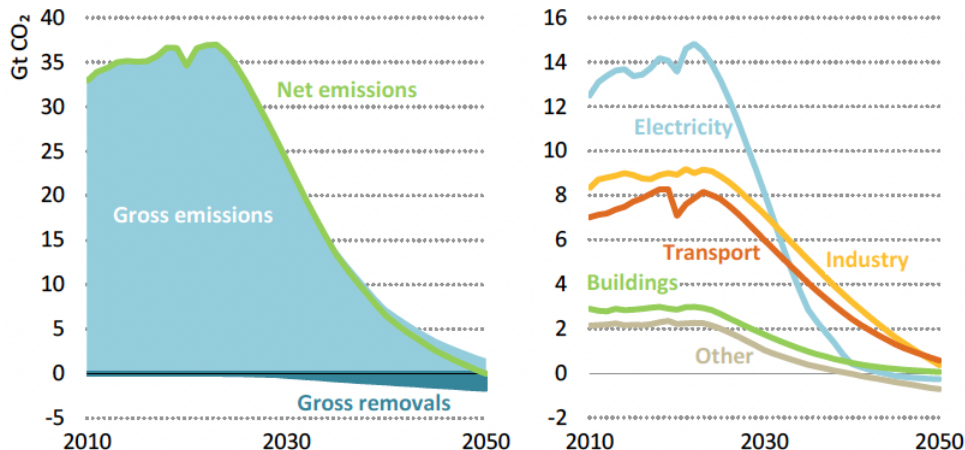
Gross emissions and removals, and net emissions by aggregated region in the NZE scenario, 2010–2050



Source: IEA (2023), Net Zero by 2050 A Roadmap – 2023 update

CO₂ Emissions in the Net-Zero Emission Scenario

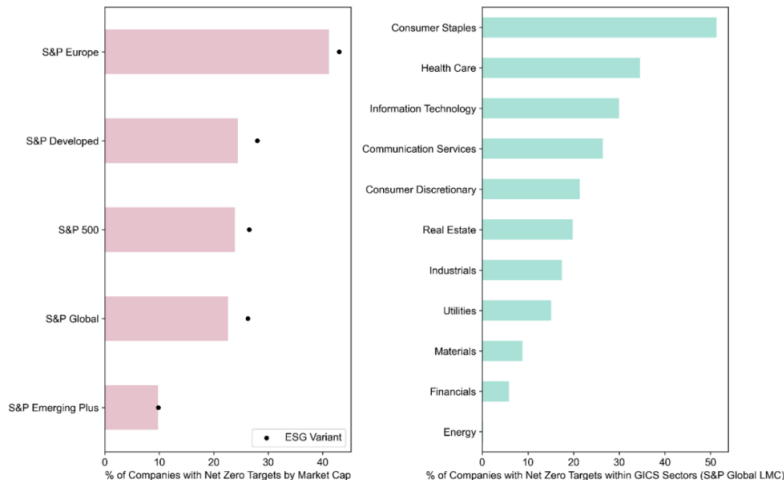
Energy sector gross emissions and removals, total net CO₂ emissions, and net emissions by sector in the NZE scenario, 2010–2050



Source: IEA (2023), Net Zero by 2050 A Roadmap – 2023 update

Regional and Sectoral-level Net Zero Pledges

Percentage of firms with net-zero targets by 2050

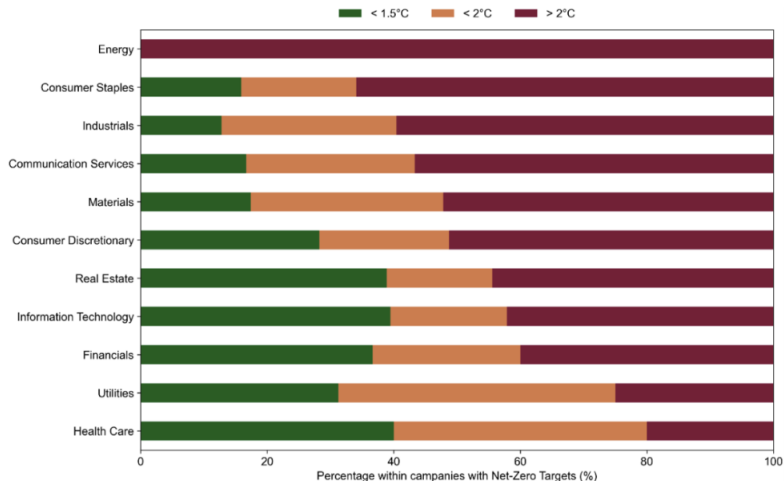


<https://www.indexologyblog.com/2023/01/31/>

[net-zero-targets-and-temperature-alignment-two-sides-of-the-same-coin/.](#)

Industry-level Net Zero Pledges

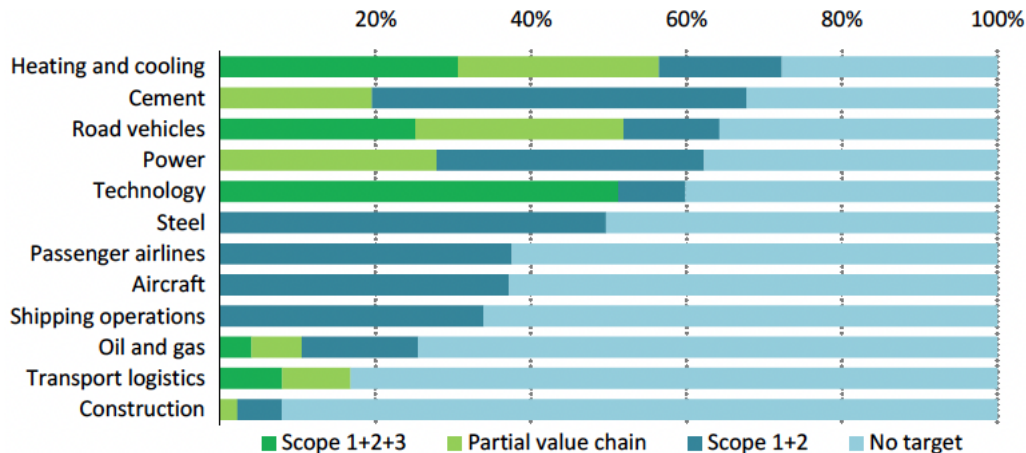
Percentage of firms with net-zero targets by 2050 (industry-level)



<https://www.indexologyblog.com/2023/01/31/net-zero-targets-and-temperature-alignment-two-sides-of-the-same-coin/>

Industry-level Net Zero Pledges

Sectoral activity of large energy-related companies with announced pledges to reach net-zero emissions by 2050



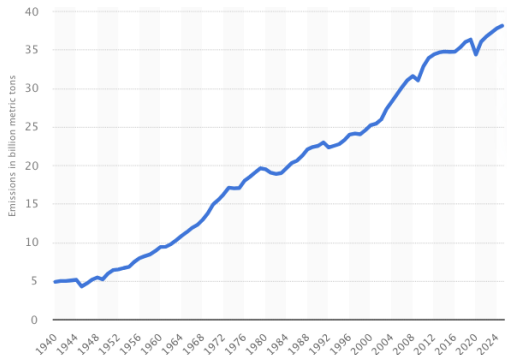
Source: IEA (2021), *Net Zero by 2050: A Roadmap for the Global Energy Sector*

Remaining Carbon Budget

IPCC (6th Assessment Report): the remaining carbon budget to limit global warming to 1.5°C is equal to 400 GtCO₂ (67% probability) (beginning of 2020)

The economic effects of Covid-19 pandemic caused fossil fuel emissions to decrease by 7% in 2020, but this effect is short lasting.

GtCO₂ emissions amount to 37.4 GtCO₂ in 2024 (Source: Statista.com)



Remaining Carbon Budget

- If the remaining carbon budget is 400 GtCO₂ and we continue to emit 37.5 GtCO₂ of carbon every year, the carbon budget will be exhausted in **2030**.
- If the remaining carbon budget is 400 GtCO₂, initial annual carbon emissions are 37.5 GtCO₂, the carbon budget will be exhausted in **2050** with a reduction of emissions by 9% per year every year.

Caveats: non-GtCO₂ emissions can increase or decrease the remaining budget by 220 GtCO₂ or more.

Net Zero versus Temperature Alignment Portfolio

- **Net-Zero Portfolio:** zero emission at a specific point in time
 - Goal: Reaching a state where the carbon "budget" of the portfolio is neutral.
 - Primary Metric: Financed Emissions (measured in tCO₂e).
 - Strategy: Often involves divestment (selling off high emitters) or offsetting to bring the total carbon footprint to zero.
 - "When": Usually anchored to a long-term target date (2040 or 2050) with interim milestones.
- **Temperature-Aligned Portfolio:** how the firms' emissions contribute to global warming
 - Goal: Ensuring the portfolio stays within a specific warming limit (1.5°C or well below 2°C).
 - Primary Metric: Implied Temperature Rise (ITR). This is expressed in degrees Celsius (e.g., "This portfolio is aligned with a 1.8°C world").
 - Strategy: Focuses on forward-looking targets. It rewards companies that have credible transition plans, even if they are currently high emitters.
 - "When": It is a constant, real-time assessment of the portfolio's current trajectory versus a scientific pathway.

Why the distinction matters

- A Net-Zero portfolio can be inconsistent with Temperature Alignment
 - A portfolio could hold a massive oil company that promises to be net-zero by 2050, but because they are not cutting emissions fast enough right now, their "implied temperature" might be 3°C.
 - Conversely, a temperature-aligned portfolio might include a "dirty" utility company that is aggressively transitioning to wind and solar, because their current actions are consistent with a 1.5°C pathway.
- **Takeaway:**
 - Net-Zero tells you where you want to end up
 - Temperature Alignment tells you if the path you are taking is fast enough to prevent a climate disaster.

Implied Temperature Rise

SAMPLE COMPANY



IMPLIED TEMPERATURE RISE **2.1°C**

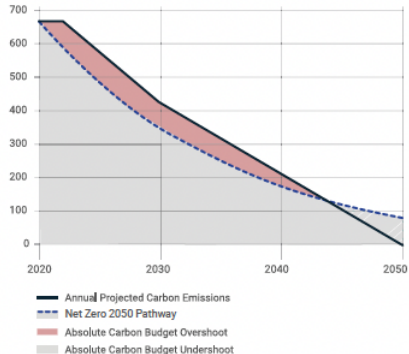
LAGGING

MISALIGNED

2°C ALIGNED

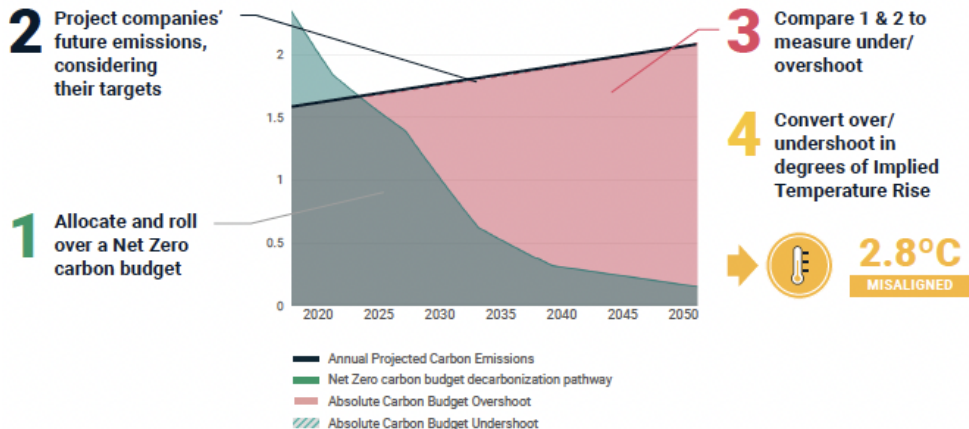
1.5°C Trajectory

Absolute emissions [Megatons CO₂e]



Source: MSCI (2024), Implied Temperature Rise. Aligning your investment strategy with a net-zero world.

Implied Temperature Rise



Source: MSCI (2024), Implied Temperature Rise. Aligning your investment strategy with a net-zero world.

Objectives of the Lecture

- Temperature Alignment and Net Zero Investment
- **Exclusion/Best-in-class Strategies**
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Jondeau, Mojon, and Pereira da Silva (2021)

Assume a reduction target θ (say 10%) of the portfolio's annual carbon emissions.

In year 1, the investor implements the exclusion or best-in-class strategy: firms with the highest carbon intensity are excluded until the overall emissions of the portfolio is reduced by a fraction θ relative to the emissions of the BAU benchmark (as observed at end of year 0)

$$CF_{Y_1}^{(p)} \leq (1 - \theta) \times CF_{Y_0}^{(b)}.$$

where

- $CF_{Y_1}^{(p)} = \frac{1}{V_{Y_1}^{(p)}} \sum_{i=1}^N o_{i,Y_1}^{(p)} E_{i,Y_0}$ (ex-ante CF based on emissions of the previous year)
- $o_{i,Y_1}^{(p)} = \frac{V_{i,Y_1}^{(p)}}{\text{Cap}_{Y_1}}$: fraction of the equity of the firm owned by the portfolio
- $V_{i,Y_1}^{(p)} = \alpha_{i,Y_1}^{(p)} V_{Y_1}^{(p)}$: dollar value invested in firm i
- $V_{Y_1}^{(p)} = \sum_{i=1}^N V_{i,Y_1}^{(p)}$: dollar value of the portfolio

How It Works

In year $n > 1$, the investor rebalances the portfolio so that the weights are again as close as possible to the weights of the BAU benchmark at the end of year $n - 1$, while reducing the carbon emissions of the portfolio by a fraction θ relative to the intensity of the portfolio at the end of year $n - 1$:

$$CF_Y^{(p)} \leq (1 - \theta)^{Y - Y_0 + 1} \times CF_{Y_0}^{(b)} \quad \text{for all } Y.$$

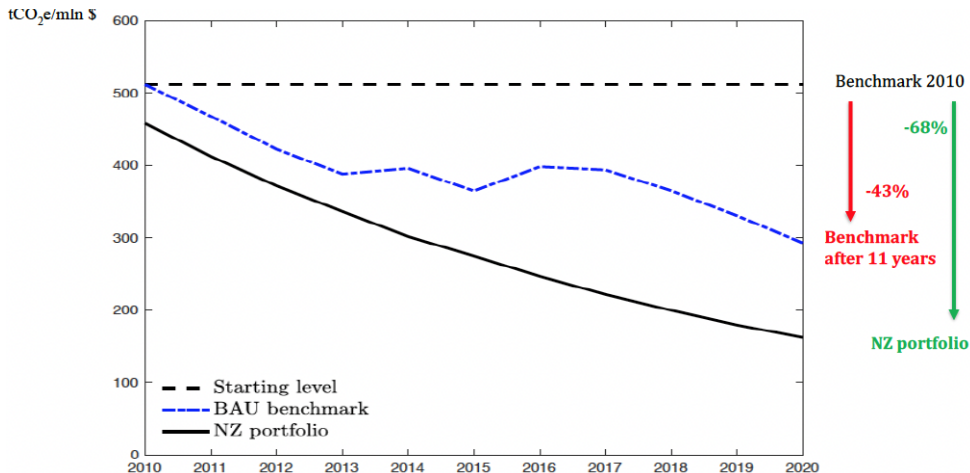
Annual emission reduction targets of $\theta = 5\%, 10\%, 15\%$ correspond to reduction over 10 years by 40%, 65.1%, and 80.3%, respectively.

EU Regulation – Article 9(3):

- Climate Transition Benchmark (CTB): reduce carbon intensity by 30% relative to the parent in year 1 and at least 7% per year thereafter
- Paris-Aligned Benchmark (PAB): reduce carbon intensity by 50% relative to the parent in year 1 and at least 7% per year thereafter

Net Zero Best-in-class Strategy

Evolution of the carbon intensity of the BAU benchmark and the NZ portfolio

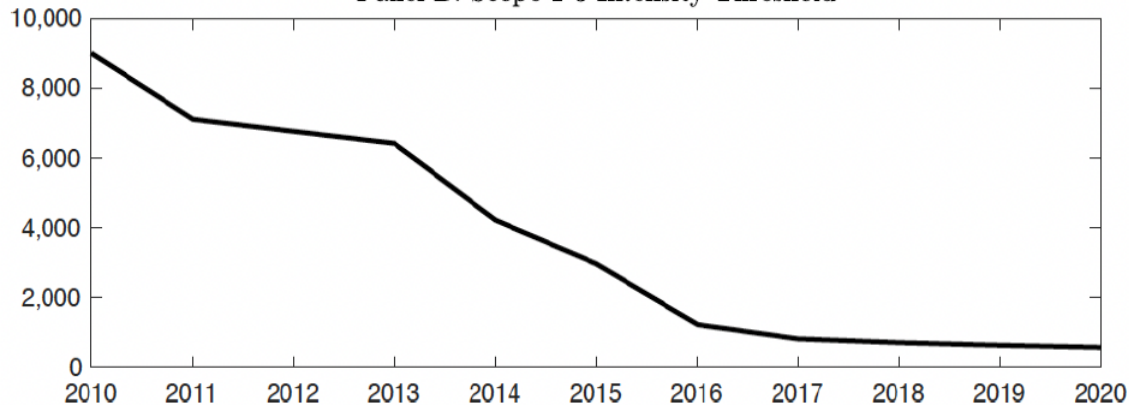


Source: Jondeau, Mojon, and Pereira da Silva (2021)

Net Zero Best-in-class Strategy

Annual carbon intensity threshold above which firms are excluded (tCO₂e/m\$ revenue)

Panel B: Scope 1-3 Intensity Threshold

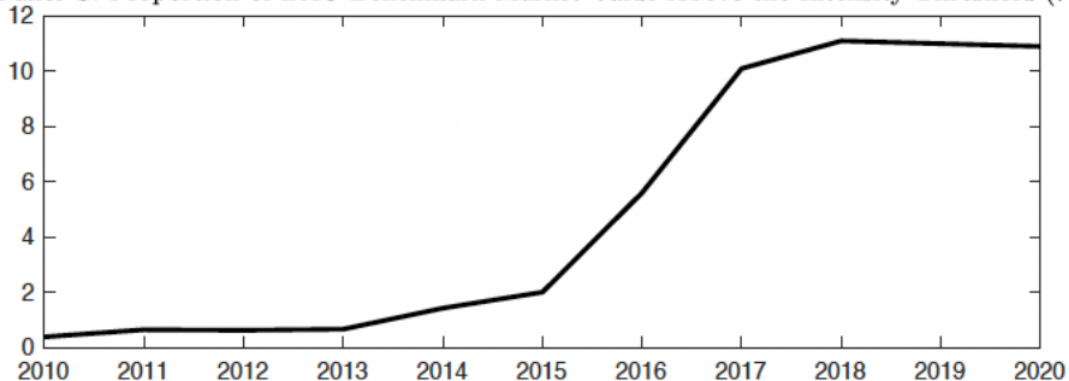


Source: Jondeau, Mojon, and Pereira da Silva (2021)

Net Zero Best-in-class Strategy

Annual carbon intensity threshold above which firms are excluded (tCO₂e/m\$ revenue)

Panel C: Proportion of BAU Benchmark Market Value Above the Intensity Threshold (%)



Source: Jondeau, Mojon, and Pereira da Silva (2021)

Net Zero Best-in-class Strategy

Dynamic reduction of the portfolio carbon footprint over 11 years (2010–2020)

	Reduction target			BAU	MSCI
	5%	10%	15%	bench.	index
Starting intensity (tCO ₂ e/\$m)	511.6	511.6	511.6	511.6	–
Final intensity (tCO ₂ e/\$m)	273.3	162.0	111.4	292.7	–
Cumulative growth (%)	-46.6	-68.3	-78.2	-42.8	–
Annual growth (%)	-5.5	-9.9	-12.9	-5.0	–
Annual return (%)	10.7	10.6	10.4	10.7	10.5
Annual volatility (%)	14.0	14.0	14.0	14.0	13.7
Sharpe ratio	0.76	0.76	0.74	0.76	0.77
Annual tracking error (%)	0.04	0.26	0.74	–	–
Number of firms excluded	28.5	210.0	463.7	–	–
Prop. of firms excluded (%)	1.1	8.0	18.1	–	–
Prop. of market value excluded (%)	0.4	5.0	13.0	–	–

Source: Jondeau, Mojon, and Pereira da Silva (2021)

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Minimization of the Tracking Error

Bolton, Kacperczyk, and Samama (2021), “Net-Zero Carbon Portfolio Alignment”

Approach 1: Minimize the tracking error of the portfolio w.r.t. the benchmark

$$\begin{aligned} \min_{\{\alpha_t^{(p)}\}} \quad & \sqrt{\text{Var}_t \left[R_{t+1}^{(p)} - R_{t+1}^{(b)} \right]} \\ \text{s.t.} \quad & \alpha_{i,t}^{(p)} = 0 \quad \text{for all } i = 1, \dots, k, \\ & \alpha_{i,t}^{(p)} \geq 0 \quad \text{for all } i = k + 1, \dots, N. \end{aligned}$$

where the k most polluting firms are excluded.

The number k is set such that the sum of carbon emissions of all the remaining constituents satisfies the carbon budget.

Minimization of the Tracking Error

Approach 2: Minimize the tracking error of the portfolio w.r.t. the benchmark

$$\begin{aligned} \min_{\{\alpha_t^{(p)}\}} \quad & TE_t^{(p)} = \sqrt{\text{Var}_t \left[R_{t+1}^{(p)} - R_{t+1}^{(b)} \right]} \\ \text{s.t.} \quad & \sum_{i=1}^N \alpha_{i,t}^{(p)} E_{i,t} \leq \bar{E}_t. \end{aligned}$$

where $\bar{E}_t$ is the carbon budget for year t to reach the net-zero carbon budget in 2050.

Minimization of the Tracking Error – Additional Features

- Tracking error: $TE_t^{(p)} = \sqrt{\left(\alpha_t^{(p)} - \alpha_t^{(b)}\right)' \Sigma_t \left(\alpha_t^{(p)} - \alpha_t^{(b)}\right)}$

where $\Sigma_t = \beta_t \Omega_t \beta_t' + \mathbf{D}_t$ is based on a factor model and $\mathbf{D}_t$ is the matrix of idiosyncratic variances.

- Constraints on sector s exposure: $\left| \sum_{i \in s} \alpha_{i,t}^{(p)} - \sum_{i \in s} \alpha_{i,t}^{(b)} \right| \leq \delta \quad (\delta = 2\%)$

Two scenarios:

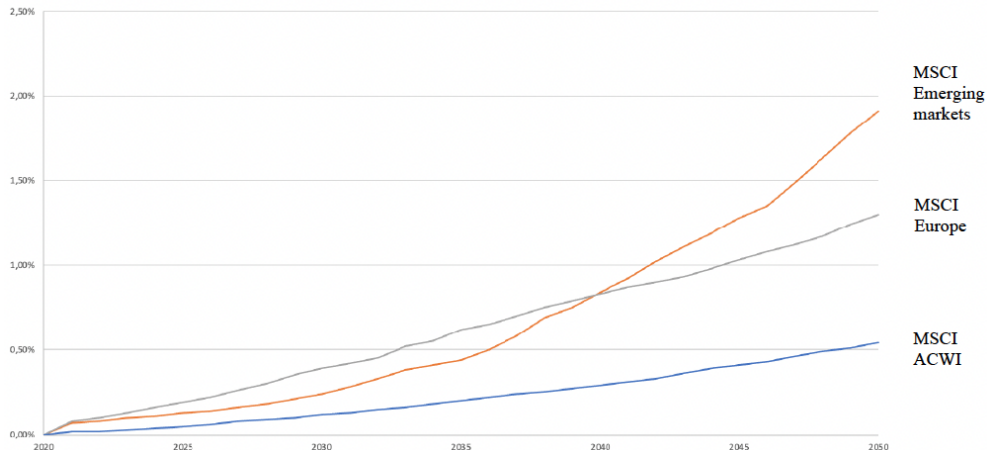
- a 25% initial reduction followed by an 8% annual reduction over 29 years
- a 10% reduction over 30 years

Remarks:

- Exclusion based on Scope 1-2-3 upstream total emissions instead of carbon intensity
- There is no constraint on regional/country exposures
- Carbon emissions are supposed constant over time

Minimization of the Tracking Error: Results

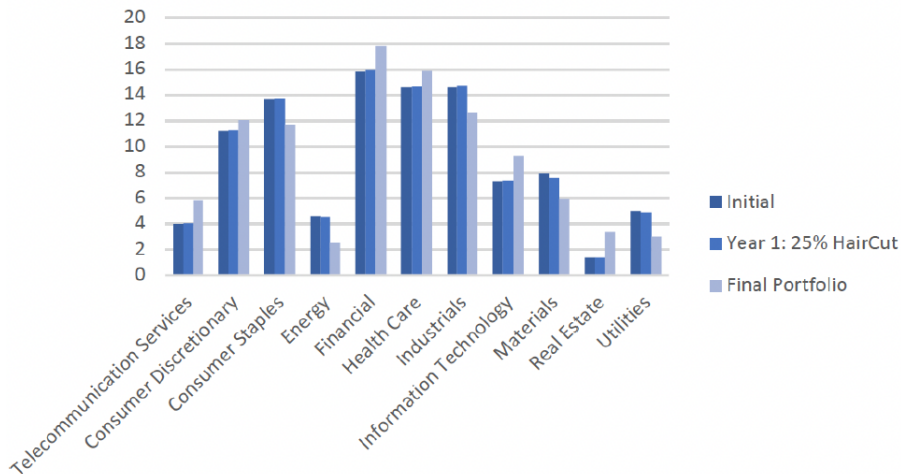
Ex-ante tracking error for MSCI indexes 1.5°C aligned portfolios



Source: Bolton, Kacperczyk, Samama (2022)

Minimization of the Tracking Error: Results

Sector deviations of the MSCI Europe 1.5°C aligned portfolio



Source: Bolton, Kacperczyk, Samama (2021)

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- **Corporate Bonds**
- Sovereign Bonds

Corporate Bonds Market

Benchmark: Bloomberg Barclays Global Aggregate Corporate Total Return Index

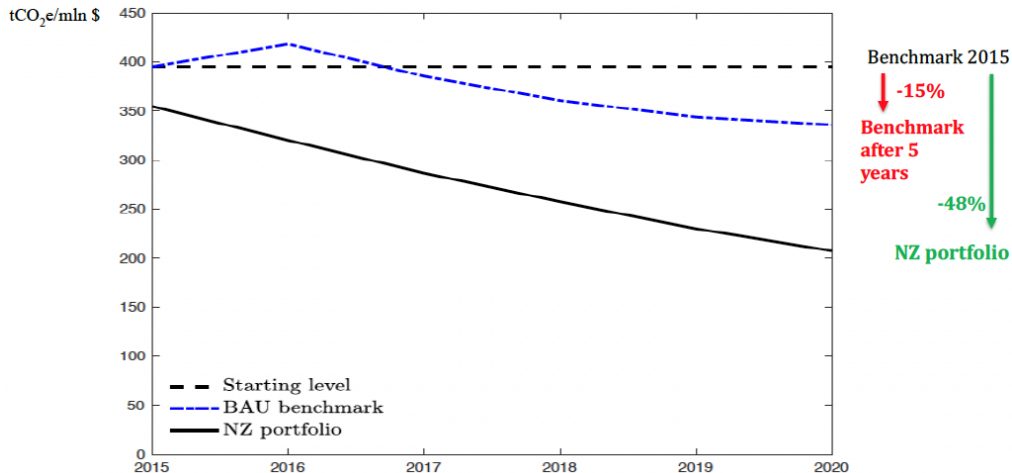
Number of constituents: 9,592 securities (759 firms) in December 2015 to 13,857 (1,331 firms) in December 2020 (11 bonds per issuer on average)

Bonds issued in USD (66.3%) and EUR (23%) and GBP (5.1%)

Bonds issued by North American firms (60%) and European firms (31%)

Financials represent 31.9% of the index on average

Evolution of the carbon intensity of the BAU benchmark and the NZ portfolio



Source: Jondeau, Mojon, and Pereira da Silva (2021)

Corporate Bonds

Dynamic reduction of the portfolio carbon footprint over 6 years (2015–2020)

	Reduction target			BAU	MSCI
	5%	10%	15%	bench.	index
Starting emissions (GtCO ₂ e)	395.2	395.2	395.2	395.2	–
Final emissions (GtCO ₂ e)	293.3	207.4	153.3	335.9	–
Cumulative growth (%)	-25.8	-47.5	-61.2	-15.0	–
Annual growth (%)	-4.8	-10.2	-14.6	-2.7	–
Annual return (%)	4.58	4.59	4.58	4.58	4.61
Annual volatility (%)	5.42	5.43	5.34	5.43	5.75
Sharpe ratio	0.85	0.84	0.86	0.84	0.80
Annual tracking error (%)	0.02	0.02	0.14	–	–
Number of firms excluded	54.8	248	553.2	–	–
Prop. of firms excluded (%)	1.06	4.22	9.12	–	–
Prop. of market value excluded (%)	0.55	3.08	7.26	–	–

Source: Jondeau, Mojon, and Pereira da Silva (2021)

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- Temperature Alignment and Net Zero Investment
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- Corporate Bonds
- **Sovereign Bonds**

Building a portfolio of sovereign bonds raises new questions:

- ① **What is the perimeter of emissions to be considered**, e.g., government or country-wide emissions?
- ② **Should emissions be based on domestic production or on domestic consumption?**
- ③ If we consider carbon intensity, **how should we normalize emissions** when production-based or consumption-based metrics are used?

(1) What is the perimeter of emissions to be considered, e.g., government or country-wide emissions?

- Only the emissions generated by the public sector's consumption of goods and services.
- Government decisions are likely to affect the entire economy, and all emissions generated within the country should be accounted for.

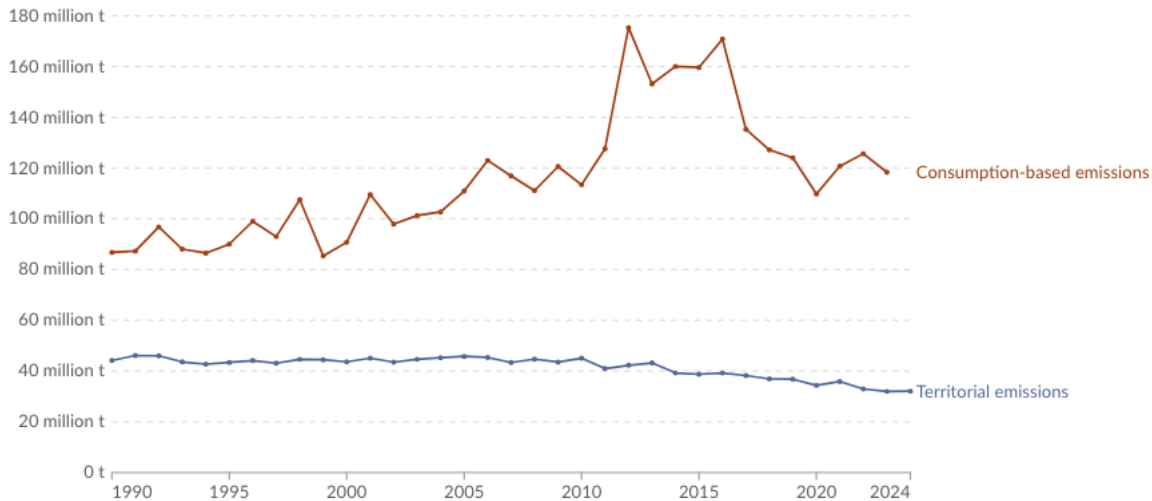
⇒ Some double counting with the private sector's emissions

⇒ But it incentivizes the government to reduce the overall country's emissions through all policy and financing tools

(2) Should emissions be based on domestic production or on domestic consumption?

- Governments tend to focus on GHG emissions in a territory (**production-based accounting**), regardless of the destination of the goods and services produced (recommended by the Kyoto Protocol). Reference for Nationally Determined Contributions (NDCs)
- It does not incorporate countries' scope 3 emissions (attributable to non-energy imports from abroad) ⇒ Carbon leakages as advanced economies tend to transfer polluting production to countries with less stringent environmental regulations
- Emissions resulting from domestic final demand, i.e., all goods and services consumed in a given country (**consumption-based accounting**): emissions embedded in imports are added to those generated from domestic consumption, while emissions from exports are excluded

Territorial and consumption-based CO₂ emissions, Switzerland



Source: Data source: Global Carbon Budget (2025),

<https://ourworldindata.org/grapher/production-vs-consumption-co2-emissions?country=~CHE>

A country's ambitious climate policies (e.g., carbon taxes) unintentionally cause GHG emissions to increase in other countries with more lenient regulations

There are two primary ways carbon leakage manifests:

- **Operational Relocation (Direct):** To avoid high carbon costs at home, a manufacturer closes its factory in the regulated country and opens a new one in a country where pollution is cheaper.
- **Trade Substitution (Indirect):** Domestic goods become more expensive because of carbon pricing. Consumers then switch to cheaper imported goods from countries that do not have those costs.

The Mechanics of Carbon Leakage: Steel Factory Scenario

Imagine a steel manufacturer in Country A and its competitor in Country B.

Step	Action in Country A (Regulated)	Reaction in Country B (Unregulated)	Global Result
1. Policy	Country A introduces a tax of \$80 per tonne of CO ₂	No carbon tax exists	–
2. Cost	Steel production costs in Country A rise by 15%	Costs remain low	–
3. Shift	The factory in A cuts production by 50% to save money	The factory in B increases production by 50% to capture the market	Zero net reduction in steel
4. Leakage	Country A's emissions drop	Country B's emissions skyrocket (less efficient tech)	Global emissions rise

How to Prevent Carbon Leakage

Governments use specific tools to plug these "leaks" so that climate progress isn't just an accounting trick.

- **CBAM (Carbon Border Adjustment Mechanism):** This is a "carbon tariff." If you want to sell steel in the EU, you have to pay a fee at the border equivalent to what an EU company would have paid in carbon taxes. This levels the playing field.
- **Free Allowances:** Under "Cap and Trade" systems, the government might give high-risk industries (like cement or chemicals) a certain amount of carbon permits for free so they are not incentivized to move abroad immediately.
- **Global Minimum Carbon Price:** If every country charged the same for carbon, there would be nowhere for the emissions to leak.

(3) If we consider carbon intensity, how should we normalize emissions when production-based or consumption-based metrics are used?

- **Production-based emissions** of a country i :

$$CI_{i,t}^{(prod)} = E_{i,t}^{(prod)} / GDP_{i,t}.$$

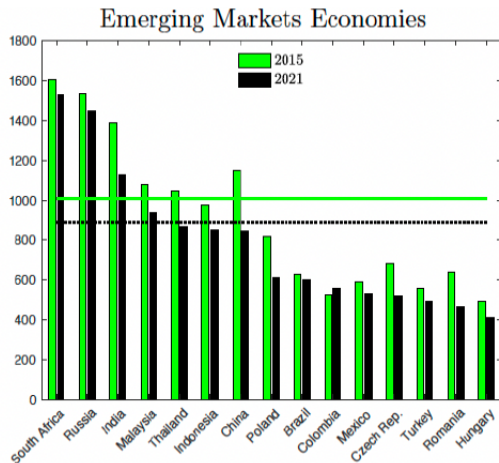
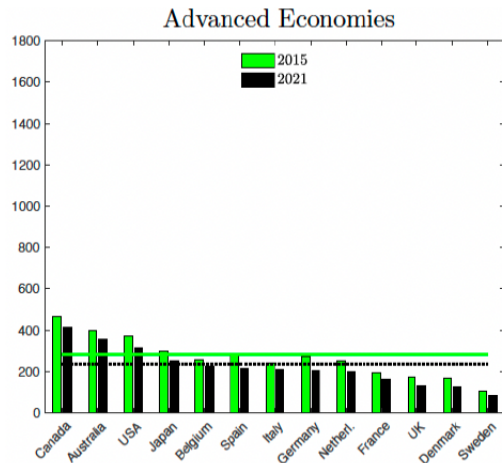
It is expressed in tonnes of CO₂ equivalent per million U.S. dollars of GDP.

- **Consumption-based emissions** of a country i :

$$CI_{i,t}^{(cons)} = E_{i,t}^{(cons)} / Pop_{i,t}.$$

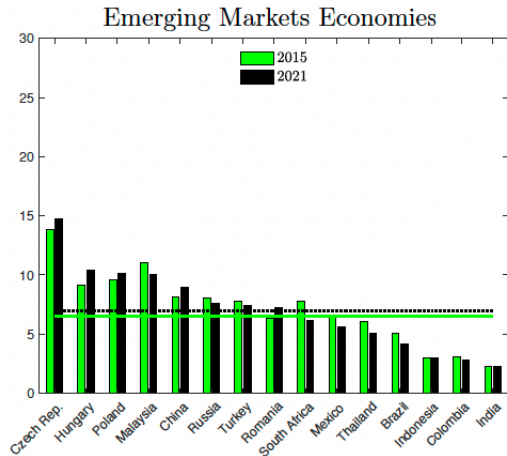
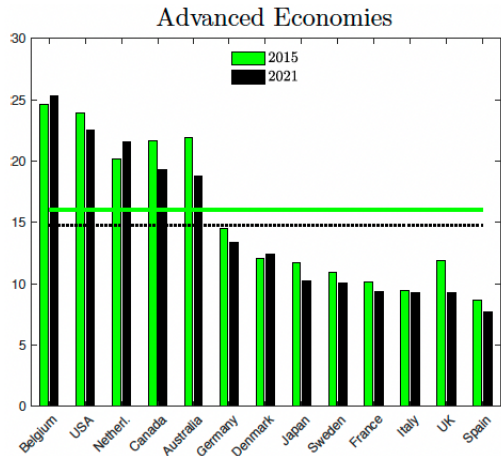
It is expressed in tonnes of CO₂ equivalent per capita.

Production-based Carbon Intensity per GDP



A production-based approach would impose most of the adjustment burden on emerging economies

Consumption-based Carbon Intensity per Capita



We use a consumption-based approach because it provides a stronger incentive to advanced economies to reduce their emissions

Cheng, Jondeau, and Mojon (2023)

Sovereign bond market

Benchmark: J.P. Morgan's Government Bond Indices (GBI)

- GBI Global: 13 advanced economies
- GBI – Emerging market Broad: 21 emerging market economies

Issues:

- ① What is the perimeter of emissions to be considered, e.g., government or country-wide emissions?
- ② Should emissions be based on domestic production or on domestic consumption?
- ③ If we consider carbon intensity, how should we normalize emissions when production-based or consumption-based metrics are used?

Objective function:

① Minimize the active share: $AS_t^{(p)} = \frac{1}{2} \sum_{i=1}^{N_t} |\alpha_{i,t}^{(p)} - \alpha_{i,t}^{(b)}|$

② Minimize the tracking error: $TE_t^{(p)} = \sqrt{\left(\alpha_t^{(p)} - \alpha_t^{(b)}\right)' \Sigma_t \left(\alpha_t^{(p)} - \alpha_t^{(b)}\right)}$

where $\Sigma_t = \text{Var}_t[R_{t+1}]$ is the sample covariance matrix of returns computed using the last 60 monthly returns.

Emissions reduction constraint:

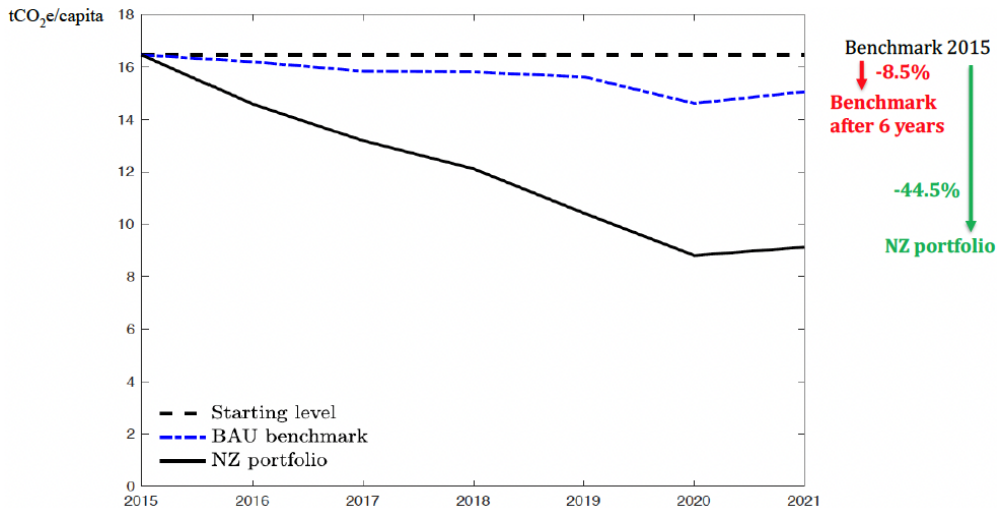
$$CI_t^{(p)} \leq CI_t^{(target)} \equiv CI_{t_0}^{(b)}(1 - \theta)^{(t-t_0+1)} \quad \text{for } t = t_0 + 1, \dots$$

where

$$CI_{t_0}^{(p)} = \sum_{i=1}^{N_t} \alpha_{i,t}^{(p)} CI_{i,t_0} \quad \text{and} \quad CI_{t_0}^{(b)} = \sum_{i=1}^{N_t} \alpha_{i,t}^{(b)} CI_{i,t_0}.$$

The NZ Portfolio Achieves Higher Carbon Intensity Reduction...

Evolution of the carbon intensity of the BAU benchmark and the NZ portfolio



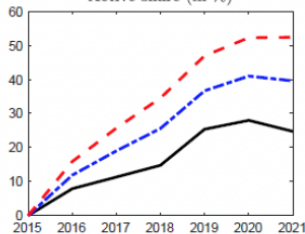
... Without Sacrificing Financial Returns

- The annualized returns of the NZ portfolio are higher (3.48%-3.57% across different annual targets) than the BAU benchmark (3.28%)
- But return volatilities are higher as measured by annual tracking error (up to 1%)
- Annual turnover is kept below 10% on average

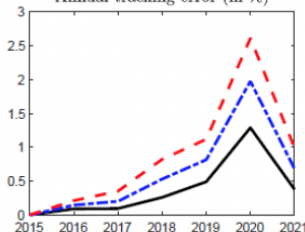
Cumulative return relative to the BAU (in %)



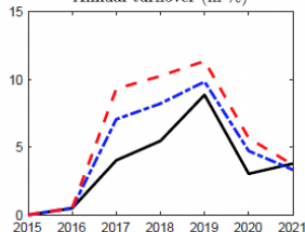
Active share (in %)



Annual tracking error (in %)



Annual turnover (in %)



... Through Radical Country Weight Rebalancing (Advanced Economies)

	BAU benchmark	NZ portfolio		
		5% reduction	7.5% reduction	10% reduction
Panel A: Advanced economies	91.07	91.07	91.07	91.07
Australia	1.44	0.03	0.00	0.00
Belgium	1.71	0.04	0.00	0.00
Canada	1.27	0.03	0.00	0.00
Denmark	0.40	1.13	1.24	1.33
France	6.85	8.99	10.56	12.01
Germany	4.76	4.00	2.60	1.61
Italy	6.34	8.92	10.93	12.81
Japan	18.38	19.51	20.04	20.49
Netherlands	1.49	0.17	0.01	0.00
Spain	3.87	6.96	9.51	11.91
Sweden	0.28	2.11	3.34	4.45
United Kingdom	6.58	8.41	9.67	10.78
United States	37.71	30.78	23.16	15.69

... Through Radical Country Weight Rebalancing (Emerging Economies)

	BAU benchmark	5% reduction	NZ portfolio 7.5% reduction	10% reduction
Panel B: Emerging economies	8.93	8.93	8.93	8.93
Argentina	0.02	0.00	0.00	0.00
Brazil	0.84	0.51	0.22	0.06
Chile	0.08	0.00	0.00	0.00
China	3.08	0.72	0.16	0.02
Colombia	0.22	1.01	1.11	1.07
Czech Republic	0.14	0.00	0.00	0.00
Dominican Republic	0.01	0.72	0.34	0.05
Hungary	0.15	0.00	0.00	0.00
India	1.39	2.45	2.93	3.26
Indonesia	0.45	1.01	0.91	0.67
Malaysia	0.24	0.00	0.00	0.00
Mexico	0.58	0.12	0.03	0.00
Peru	0.09	1.00	1.23	1.31
Philippines	0.01	1.32	1.97	2.48

Unconstrained Rebalancing Entails Several Issues

Operational issues

- High-quality sovereign debt is scarce resource → hard to exclude
- Small countries with large emission reduction → hard to increase supply
- Technically difficult to divest largely from dollar assets

Rebalancing may also induce **macro-financial risks**

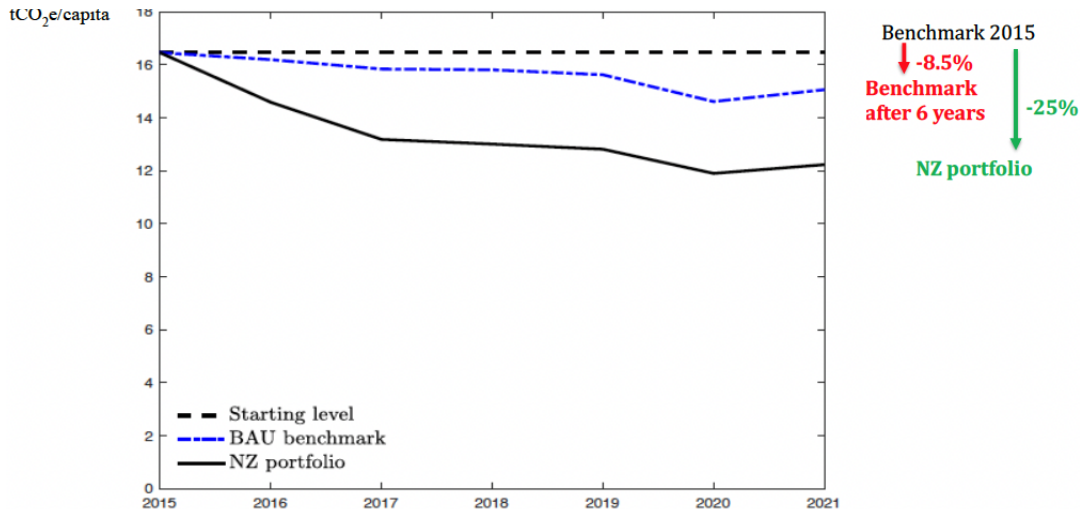
- Divested countries → increase in funding costs
- Rebalancing between currencies → exchange rate fluctuations

We set **boundaries for the portfolio rebalancing** (constrained approach)

- To keep the same weights for AE and EM as in BAU benchmark
- To keep the creditworthiness of the portfolio unchanged
- To limit country weights between 0.5 and $1.5 \times$ BAU benchmark weights

Constrained Approach

Evolution of the carbon intensity of the BAU benchmark and the NZ portfolio



End of Lecture 9